

Fig. 6: Actual-size PCB layout of the robot section

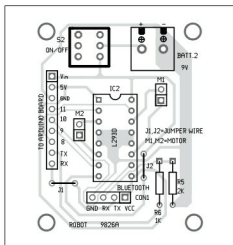


Fig. 7: Component layout of the PCB shown in Fig. 6

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

Arduino UNO is programmed using Arduino IDE software. ATmega328 on Arduino UNO comes with a bootloader that allows you to upload new code to it without using any external hardware programmer. It uses STK500 protocol to communicate. You can bypass the bootloader and program the MCU through in-circuit serial programming (ICSP) header, but using bootloader programming is quick and easy. Select the correct board from Tools Board menu in Arduino IDE and burn the program (sketch) through a standard USB port in the computer.

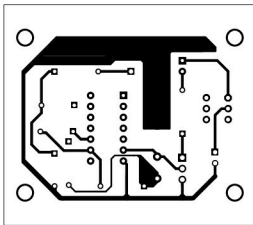


Fig. 8: Actual-size PCB layout for the detector section

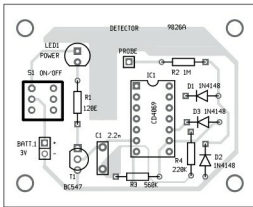


Fig. 9: Component layout of the PCB shown in Fig. 8

SENA BTerm Bluetooth Terminal is the Android app used in this project. It is available for free on Google Play.

Construction and testing

An actual-size PCB layout of the robot section is shown in Fig. 6 and its component layout in Fig. 7. An actual-size PCB layout for the detector section is shown in Fig. 8 and its component layout in Fig. 9.

The whole circuit can be accommodated in a small PVC pipe and used as a handy broken-wire detector.

Before using this robot for detecting broken wires, first find out the faulty or broken wire by continuity method using any multimeter or continuity tester. Then connect 230V AC mains live wire at one end of the faulty wire, leaving the other end free. Connect neutral terminal of the AC mains to the remaining wires at one end.

For single-wire testing, connecting neutral only to the live wire at one end is sufficient to detect the breakage point. In this circuit, a 10cm long, thick, single-strand wire is used as the test probe. To detect the breakage point, turn on switch S1 and slowly move the test probe close to the faulty wire. LED1 will glow if AC voltage is present in the faulty wire; it goes off if there is a breakage point in the wire. The point where LED1 turns off is the broken-wire point. While testing a broken 3-core rounded cable wire, bend the probe's end in the shape of J and keep the test probe close to the cable to increase its sensitivity. During testing, avoid any strong electric fields near the circuit to prevent false detection.

Controlling the robot

To control the robot using your Android phone, first install SENA BTerm app. Following steps are as per Lenovo smartphone.

After installation, you will see SENA icon on your device. Click on it and select Bluetooth Management. Select Connect To→Select→Start Scan. It will detect nearby Bluetooth devices.

In this case, the Bluetooth device detected by your phone will be HC-05. Select it from the list. With this, your phone will connect to JY MCU Bluetooth module on the robot. Five different keys on the phone are used to control the robot, namely, w for forwards, s for backwards, a for left turn, d for right turn and x for stop. If you select w on your smartphone, a 'forward' message will be displayed on your phone screen, and at the same time the robot will move forwards and continue until you press x. ●



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